Electric Power Systems Research 183 (2020) 106278

Contents lists available at ScienceDirect

Electric Power Systems Research

journal homepage: www.elsevier.com/locate/epsr

A converging non-cooperative & cooperative game theory approach for stabilizing peer-to-peer electricity trading☆

Waqas Amin[a], Qi Huang*[,a], M. Afzal[a], Abdullah Aman Khan[b], Khalid Umer[a], Syed Adrees Ahmed[a]

[a] Sichuan Provincial Key Lab of Power System Wide-Area Measurement and Control, School of Mechanical and Electrical Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China

[b] School of Computer Science and Engineering, University of Electronic Science and Technology of China

ARTICLE INFO

Keywords:
P2P electricity trading
Power market
Prosumer
Game theory
Cooperative game
Non-cooperative game

ABSTRACT

P2P electricity trading is a promising new approach, which achieves a higher level of satisfaction for prosumers and consumers, and reduces the amount of line losses in smart-grid. However, uncertainties in demand and supply of energy may result in contract instability for prosumers and consumers in P2P market. This paper proposes a framework which incorporates both non-cooperative and cooperative game, to facilitate the P2P electricity trading while maintaining stability of the contract. The proposed cooperative game ensures the survival of small-scale prosumers at a risk of market-exit when they do not own enough surplus electricity to cater the power needs of a single consumer. Hence the stability of the contract can be guaranteed. In the proposed approach, a pivotal player is identified with the concept of Shapely-Shubik power index, to distribute the shared revenue in a fair manner. An IEEE 14-bus system, which includes total 22 players (11 prosumers and 11 consumers), is used to test the effectiveness of the proposed approach. One year of dataset is tested and it is shown that both prosumers and consumers can get benefit from the P2P electricity trading while maintaining the stability of the market.

1. Introduction

With the increase in penetration of renewable energy resources such as rooftop photovoltaic (PV) panels, small wind turbines along with improved information and communication technology (ICT) devices, traditional energy consumers can act as prosumers, which can both utilize and sell energy [1,2]. Due to the stochastic nature of DERs, it is usually difficult to predict the demand and supply of electricity. Consequently, when prosumers have surplus electricity, they store in batteries and sell to other consumers or to the grid. The direct energy trading between prosumers and consumers is known as peer-to-peer (P2P) energy trading, which usually invokes within a local electricity distribution system. In P2P energy trading, peer is generally a single entity or a group of entities, that can trade electricity directly with any other peers without involving any intermediations e.g. local energy suppliers [3]. In smart grids P2P energy trading is usually facilitated by ICT based services [4]. In unidirectional conventional energy trading model, electricity is normally transmitted from largeâscale generators to consumers over long distances by means of electricity transmission networks, whereas cash flows follow the opposite way. In contrast, the P2P energy trading market enables multidirectional trading within local P2P market. Many trials have already been carried out for implementing the energy trading based on the ”P2P economy” for example, sonnenCommunity in Germany [5], Piclo in UK [6] and Vandebron in the Netherlands [7]. These trials focused on providing incentives and tariffs to electricity customers from the energy suppliers perspective. The role of P2P energy trading at the level of distribution network has also been investigated in recent years. In [8], a P2P based network method was proposed for saving the energy bills for all micro-grids and improving the utilities of local DERs. Alam et al. [9], proposed an integrated demand-side management system coordinated with P2P energy trading to reduce the energy cost in smart grid. Afzal et al. [36], proposed an distributed demand-side management system coordinated with community energy trading, to reduce the individual home energy bill in community microgrid with the use of Bc to provid trusted communction between buyers and sellers. In [10], a price based demand response energy sharing model was proposed. Furthermore, role of multi-agent energy management system in micro-grids has been widely discussed in recent years. Roche et al. [11] presented overall review of several multi-agent systems that were used for grid energy management. Game theory is a mathematical tool that helps in analyzing the interactions of prosumers and consumers considering them as market players, in the form of game. Several studies have been carried out to address the interaction of prosumers and consumers in form of a contract. The proposed

☆ This Paper is supported by Sichuan Youth Science and Technology Innovation Team Fund (grant no. 2017TD0009).
* Corresponding author.
E-mail address: hwong@uestc.edu.cn (Q. Huang).

https://doi.org/10.1016/j.epsr.2020.106278
Received 6 July 2019; Received in revised form 2 January 2020; Accepted 18 February 2020
Available online 06 March 2020

methods in [12–14] addressed the contracts, which enabled the electricity supply to hedge unfavorable low market prices and financial losses caused by the mismatch between supply and demand in the spot market. Dai Yu et al. [15], proposed an electricity supply chain coordination model based on quantity discount contract, considering only fixed demand load. Nagurney et al. [16] and Wu et al. [17] proposed a static electric power supply chain and a network equilibrium model respectively. In [18], a non-cooperative game-based competition model between demand and selling energy was illustrated. In [19], retailing spot market of electric energy based on cooperative multi-agent system was proposed. A novel hourly-ahead profit model for an active distribution company with high level of connected distributed generators, that makes sale proposals for the pool-based system market, was proposed in [20]. The participation of different energy prosumers as a single entity to reduce the total energy cost through the forecasting inaccuracies has been discussed in [21]. Considering the related uncertainties, [22] proposed a method based on Nokaido-Isoda in electricity market, by which DERs were able to maximize their profit through the price bidding strategy. A non-cooperative game between multiple utility companies and a revenue model that considers the load forecast error was proposed in [23]. In [24], a model based on the deterministic and stochastic nature of deregulated electric power market was proposed. In [25], a non-cooperative game framework devises an optimum bidding strategy for prosumers, on the basis of historical electricity market data. Wang et al. [26], proposed an energy trading game with incomplete information. A cooperative game with punishment mechanism was developed in [27]. An incentive-based contractual game considering different prosumers, which are categorized based on unit production cost was proposed in [28]. However, the author proposed game only addresses one consumer, which seems to be impractical, as in reality, P2P accommodates more than one consumer. The previous research only focuses on the interaction of prosumers and consumers, either in cooperative or non- cooperative games. However, in all the bidding strategies and pricing mechanism, there is a high probability that some prosumer or consumer may face an exit situation, either because of not getting matched entities or not getting the expected price from the market. This may lead to a monopolized market, in which not all players are able to trade in a fair manner, leading to uncertainties and contract instabilities. It is therefore imperative to find such a solution, that will not only encourage fair competition among the players but also appeal both the prosumers and consumers to become a part of P2P electricity trading market. To address these gaps, the authors propose a game theory-based mechanism, which ensures the positive utility for both the trading entities i.e. prosumers (profit) and consumer (savings in bill). The proposed mechanism considers both non-cooperative and cooperative games simultaneously. The main contributions of this paper are as follows:

1. This paper proposes a framework in which the P2P players can interact with each other to make a profit while maintaining the stability of the market.
2. The change-over mechanism from the non-cooperative game model to cooperative game model is proposed. The proposed model automatically converges from non-cooperative game model to the cooperative game.
3. A Shapely based method, which distributes the shared revenue among the prosumers in a fair manner, is designed.

The rest of the paper is organized as follows. Section II describes the system model. The consumer and prosumer models are further discussed in section III. In section IV, the discussion is extended towards a non-cooperative game model followed by the cooperative game model. Section V discusses identification of pivotal player and distribution of shared revenue. The experiment results are shown in Section VI and the conclusion is presented in Section VII.

2. System model

The proposed P2P market model assumes, that there are ϕ_n peers interacting with each other at any time t. Among ϕ_n peers there are ϕ_s and ϕ_b peers are acting as consumers and prosumers respectively in such a way that: $\phi_s, \phi_b \epsilon \phi_n$ *and* $\phi_s \cap \phi_b = \phi$, Where ϕ presents an empty set. The proposed P2P electricity market consists of a community manager system (CM), which manages and monitors all the trading activities inside a community. In rest of the paper, we address community manager as a system. Generally, the P2P electricity trading market is based on peers that share common interests and goals i.e. electricity trading and positive utility. At time t, the consumer informs the system about his electricity requirements Q_{demand}, along with the willingness factor C_w. The factor C_w helps the system to distinguish between the consumers willing to participate in P2P market ($C_w = 1$) with the consumers which intended to import energy directly from the grid ($C_w = 0$). Similarly, a prosumer will also informs the system about the amount of surplus energy it has ($Q_{surplus}$), along with the willingness factor P_w. Moreover, P_w helps the system to manage and separate prosumers into groups. ($P_w = 1$) represents the prosumers which are willing to participate in P2P market; whereas, ($P_w = 0$) represents the prosumers preferring to export energy to the grid. In the case of mismatch between demand and supply, the energy can be imported from or exported to the grid. Moreover, it is assumed that at any time t, a prosumer either sells surplus electricity to the P2P market or exports it to the grid. The system will charge a small amount as a fee on each successful contract formulation.

A general mathematical formulation of the proposed market can be written as Moret and Pinson [29] (Fig. 1):

$$\min_{D} \sum_{n \epsilon \phi} C_n(p_{n,} q_{n,} C_{w,} P_{w,}) + G(e_{imp,} e_{exp}) \tag{1}$$

$$s.\,t\ p_n + q_n + C_w + P_w = 0,\ \forall\ n\ \epsilon\ \phi_n \tag{2}$$

$$\sum_{n \epsilon \phi_b} q_n = 0 \tag{3}$$

$$\sum_{n \epsilon \phi_b} C_w = e_{imp} \tag{4}$$

$$\sum_{n \epsilon \phi_s} P_w = e_{exp} \tag{5}$$

Where $D = (p_n, q_n, C_w, P_w)_{n \epsilon \phi}$ is the set of decision variables. p_n represents the surplus energy of the prosumers, q_n is the quantity required by the consumers. C_w, P_w represents the decision variables for consumers and prosumers for trading energy either in the P2P energy market or with grid. C_n is the representation of cost associated with the decision variable. And $G(e_{imp}, e_{exp})$ is the function associated to the energy exchange with outside P2P energy market.i.e.(grid).

The constrains in Equ.(2) describes the power balance in P2P energy market i.e.(net generation and energy traded). This energy can be traded either in P2P energy market (3) or it can be traded with the grid (4-5).

3. Consumer and prosumer model

3.1. Prosumers model

Considering the game theory framework, prosumer earns a utility U after selling the surplus electricity $Q_{surplus}$ at time t. All prosumers are strictly rational and aware that selling surplus electricity to the P2P electricity market can benefit more, as compared to the grid [30–33].

$$U_{pro_{p2p}} > U_{pro_{Grid}} \tag{6}$$

Moreover, prosumers are highly motivated to participate in P2P electricity market. The proposed model restricts a prosumer P_i to trade electricity lower than a certain amount (ξ). By limiting the amount to (ξ), the proposed model gives equal opportunity to trade among multiple consumers rather with a single consumer having a larger electricity demand. Moreover, this limitation (ξ) attracts consumer to participate in P2P electricity market more frequently, which guarantees

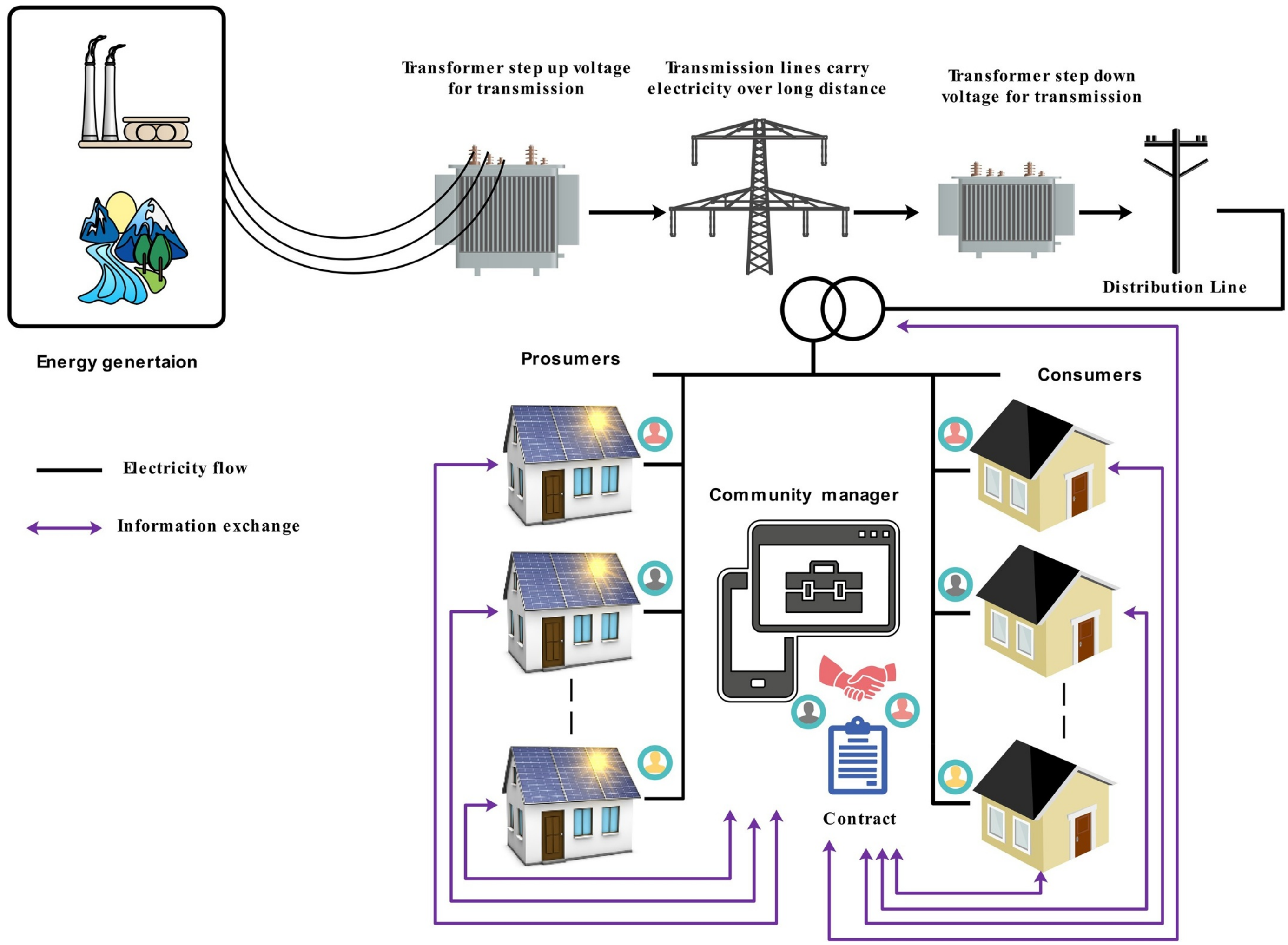


Fig. 1. Overview of the market model.

reduction in electricity bills. if u_1, u_2, u_3, ...u_n are utilities of a prosumer P_i which he/she gets after selling the surplus electricity at price p_1, p_2, p_3, ...p_n to the consumers c_1, c_2, c_3, ..., c_n then the revenue function R_i of a prosumer P_i can be written as $R_i = (\sum_{i=1}^{n} q_i{}^*p_i)$. The objective function of a prosumer can be formulated as:

$$\max_{i\epsilon\phi_s} \sum_{i=1}^{n} R_i = \max \sum_{i=1}^{n} (q_i{}^*p_i) \tag{7}$$

$$s.\ t\ q_i \le \xi \tag{8}$$

where q_i represents the total surplus electricity of a prosumer p_i. If q_i' represents the trading quantity of a prosumer P_i towards a consumer c_i then $q_i^* = q_i - q_i^{'}$ is the remaining surplus electricity of a prosumer. However, if $q_i^* \neq 0$, then the constraints in (8) will not apply and prosumer P_i can trade with the same consumer c_i if system meets the following condition:

$$Q_{demand,-i} = 0\ and\ Q_{surplus,-i} = 0 \tag{9}$$

where $-\ i$ represents the consumers or prosumers other than i.

3.2. Consumer's model

A consumer needs Q_{demand} of electricity at time t. On receiving the required amount of electricity, the consumer pays the bill to the prosumer. Typically, the price of electricity offered by the prosumers in P2P electricity market is comparatively lower than the price of imported electricity from the grid [30–33]. Therefore, consumers are highly motivated to participate in P2P electricity market which consequently, increases the participation of consumers in P2P market.i.e. $U_{conP2P} > U_{conGrid}$. Furthermore, If q_1, q_2, q_3, ...q_n are the purchased units of consumer c_i, from the prosumers P_1, P_2, P_3, ...P_n after paying an amount p_1, p_2, p_3, ...p_n, then the total bill for the consumer c_i can be formulated as: $B_{iBill} = (\sum_{i=1}^{n} q_i{}^*P_i)$, the objective function for the consumer c_i can be written as:

$$\min_{i\epsilon\phi_b} \sum_{n}^{i=1} B_{i_{Bill}} = \min\left(\sum_{i=1}^{n} q_i{}^*p_i \right) \tag{10}$$

$$s.\ t\ q_i''\epsilon\ c_i \leqslant \xi \tag{11}$$

where q_i'' represents the total electricity required by a consumer c_i. If $\check{q}_i$ represents the trading quantity of a consumer c_i towards a prosumer P_i then $q_i^{**} = q^{''} - \check{q}_i$ represents the remaining electricity demand of a consumer c_i. However, if $q_i^{**} \neq 0$, then the constraints in (11) will not apply and consumer c_i can trade with the same prosumer P_i if system meets the following condition:

$$Q_{demand,-i} = 0\ and\ Q_{surplus,-i} = 0 \tag{12}$$

4. Contract formulation among consumer and prosumer

4.1. Non-cooperative game formulation

The proposed game starts at time t, the system will initiate a pricing signal P_{market} to start the game and in response, each consumer submits the information including the required electricity quantity q_{req}, the offered price

P_{offer} and eagerness factor δ to the system. The eagerness factor δ tells the system, how much a consumer is in need of electricity. Additionally, the value of δ can range between $0 \sim 1$. Likewise, each prosumer submits the quantity of surplus electricity $q_{surplus}$ and the proposed price $p_{propose}$ to the system. Prosumers and consumers submit the price based on the following criteria to select the trading price p_{trad}in a contract as discussed in [34].

$$p_{propose} \leq p_{market} \tag{13}$$

$$p_{offer} \geq p_{market} \tag{14}$$

$$p_{trade} = average(p_{propose}, p_{offer}) \tag{15}$$

Eq. (15) describes the trading price among the trading pair i.e.(consumer, prosumer) which is selected as the average of the prices a prosumer proposed to the consumer offered. After gathering this information from the players, all the information excluding δ, is broadcasted to all the prosumers. The reason to omit δ is to restrict a prosumer to take advantage of the consumer eagerness. However, to avoid false reporting of the consumer's eagerness, model proposes to impose some extra charges other than the contract formulation charges to all those consumers report a high eagerness factor i.e. $\delta > 0.8 \sim 1$. By doing this we are restricting consumers about false reporting about the consumer's eagerness factor. Formally, a strategic-form game Ξ is a tuple as $\Xi = \langle N, A, U \rangle$, where, N is the finite number of players which contains prosumer $P = \{P_1, P_2, P_3, ...P_n, \}$ and consumer $c = \{c_1, c_2, c_3, ...c_n, \}$, as disjoint set such that $P_i \cap c_i = 0$ Moreover, A is the action profile of players and it contains $\{P_{offer}, P_{propose}\}$, as consumers' and prosumers' action respectively. $U: A_1 \times ... \times A_n \rightarrow R$ is the i^{th} players' utility. Each prosumer $P_i \epsilon P$ has utility match u_{ij}^P with consumer $c_j \epsilon\, c \cup \phi$, where ϕ represents no match. Similarly, for each consumer $c_j \epsilon\, c$, u_{ij}^c is the match utility from matching to prosumer $P_i \epsilon P \cup \phi$. We denote $U^p = (u_{ij}^P)_{i \epsilon P, j \epsilon c}$ as prosumer consumer pair for contract. Moreover, a strict preference for a prosumer P_i, $u_{ij}^P \neq u_{ij'}^P$ for any $j, j' \epsilon\, c \cup \phi$ and for any consumer c_j, $u_{ij}^c \neq u_{i'j}^c$ for any $i, i' \epsilon P \cup \phi$; match utilities are strictly positive for all $P_i \epsilon P$ and $c_j \epsilon c$, $u_{ij}^P > 0$ and for all $c_j \epsilon c$ and $P_i \epsilon P$, $u_{ij'}^c > 0$, all consumers and prosumers are prefer to be matched over remaining unmatched Therefore for any $P_i \epsilon P$. All consumers $c_j \epsilon c$ are acceptable where, $u_{ij}^P > u_{i\phi}^P$ and for any $c_j \epsilon c$ all prosumers $P_i \epsilon P$ are acceptable where, $u_{ij}^c > u_{\phi j}^c$. Based on the information provided by the system, all prosumers assign a rank to the consumers and make their own preference list of consumers, with whom they want to make a contract for trading. Considering all prosumers are rational a prosumer P_i played his strategy $\left(p^*_{i_{propose}}\right)$ as best response to the other prosumer's P_{-i} played strategies $\left(p_{-i_{propose}}\right)$ such that: $p^*_{i_{propose}} \epsilon\, BR\left(p_{-i_{propose}}\right) if\ \forall_i\ , u_i\left(p_{i_{propose}}, p_{-i_{propose}}\right) \geq u_{-i}\left(p_{i_{propse}, p_{-i_{propose}}}\right)$. All prosumers prefer to trade first with consumers offering a higher price. If price offered $\left(P_{c_{i_{offer}}}\right)$ by the consumer c_i is higher, such that $p_{c_{i_{offer}}} > p_{c_{-i_{offer}}}$, the prosumer ranks the consumer in his preference list as $c_i \succ c_{-i}$. Although, different consumers based on different eagerness factor δ, rank different prosumers in their preference list. Once the preference lists from both side of the players (consumers and prosumers) are ready, the lists are sent back to the system. Afterwards, the system establishes a contract between prosumer and consumer based on the Algorithm 1.

As a result, a contracted pair for each consumer and prosumer can now trade the electricity in P2P market w.r.t constraints as shown in (8) and (12). If a consumer requires energy exceeding ξ or a prosumer has energy more than ξ then, the system will follow the Algorithm 2 to fulfill their requirements. The system will follow the algorithms 1 & 2 to form the contracts (one-to-one) between consumers and prosumers as long as the following situation occurs:

$$q_{i_{required}} = 0 | q_{i_{surplus}} = 0 | q_{i_{surplus}} < q_{i_{required}} \tag{16}$$

In the case,$q_{i_{surplus}} < q_{i_{required}}$, the proposed model converges from non-cooperative game model to cooperative game model.

Algorithm 1 : contract formulation for ξ KWh trading of electricity

1: *Initialize all $P_i \epsilon P$ and $c_i\ \epsilon\ C$. Where, P represents the values of the surplus energy of all prosumers and C represents the values for demand of energy of all the consumers*
2: *While consumer c_i has prosumer P_i to propose*
3: c_i = first consumer on P's list to whom P_i has not yet proposed
4: if c_i is free
5: (P_i, c_i) become engaged
6: elseif : some pair (P_i', c_i) already exists
7: if c_i prefers P_i to P_i'
8: P_i' becomes free
9: (P_i, c_i) become engaged
10: else
11: (P_i', c_i) remain engaged

Algorithm 1 :contract formulation (After ξ KWh trading of electricity)

1: *Initialize all $P_i \epsilon P$ and $c_i\ \epsilon\ C$. Where, P presents the values of surplus energy left after 1st contract of all prosumers and C presents the values for the demand of energy of all the consumers left after 1st contract*
2: *While consumer c_i has prosumer P_i to propose*
3: c_i = first consumer on P's list to whom P_i has not yet contracted
4: if c_i is free
5: (P_i, c_i) become engaged
6: elseif : some pair (P_i', c_i) already exists
7: if c_i prefers P_i to P_i'
8: P_i' becomes free
9: (P_i, c_i) become engaged
10: else
11: (P_i', c_i) remain engaged

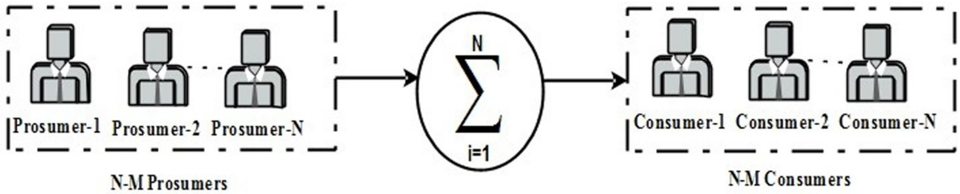


Fig. 2. Cooperative Game Formulation.

4.2. Cooperative game formulation

The proposed system also deals with the situation in which a prosumer P_i is left un-contracted, due to not having enough surplus electricity to fulfill the electricity requirement of any consumer c_i, as presented in (16). To overcome this problem, a stage-hunt game model is employed to form a grand coalition as shown in Fig. 2. If N is the total number of prosumers that participate in non-cooperative game then, $(N - M)$ presents such prosumers still having surplus energy but not in such quantity that they can meet the requirement of a single consumer. Similarly, $(N - M)$ represents such types of consumers, still requiring energy after trading in non-cooperative game.

Formally, a strategic-form game Ξ' is a tuple, i.e. $\Xi' = (P', (S_{i,\epsilon P'}), (u_i)_{i\epsilon P'})$ where, P' is the set of all prosumers, having surplus electricity less than the consumer demand i.e. $\left(q_{i_{surplus}} < q_{i_{required}}\right)$ So, $P' = \left\{q_{1_{surplus}}, q_{2_{surplus}}, q_{3_{surplus}}, \ldots, q_{n_{surplus}}\right\}$ where S is the strategies, $(P_{propose})$ is the set of all prosumers in such a way that, $S = \left\{p_{1_{propose}}, p_{2_{propose}}, p_{3_{propose}}, \ldots, p_{n_{propose}}\right\}$ and U is the utility function of the prosumers $U: S \to R$. According to the proposed grand coalition, each prosumer has only two options as strategy adopting (strategy A) or defecting (strategy B). While following strategy (A), any prosumer being a part of grand coalition is assumed committed to the trading price in (17), rather than their own proposed price.

$$p_{trade} = average(P_{1_{propose}}, P_{2_{propose}}, P_{3_{propose}}, \ldots, P_{n_{propose}}) \tag{17}$$

The utility u_i of any prosumer P_i can be expressed as:

$$u_i = \begin{Bmatrix} \beta_i(s) - \gamma_i, & s_i = A \\ 0, & s_i = D \end{Bmatrix} \tag{18}$$

where $\beta_i(s) - \gamma_i$ is the net profit of being a part of grand coalition and γ_i is the grand coalition formulation cost which is considered negligible in this case. It can be said that, Ξ' is a stag-hunt game if, for all $P_i \epsilon P'$, $S_i = \{A, D\}$ i.e., all prosumers have common or single strategy either A or D and the payoff function of the grand coalition is an increasing function of the prosumers who are a part of grand coalitions and stay in P2P market for contract with consumers, whereas the payoff of choosing D is a constant say x. Furthermore, suppose A^n represents the profile in which every prosumer is a part of grand coalition that hunts deer, we have that, for all $P_i \epsilon P'$, $u_i(A^n) > x$. Moreover, any unilateral deviation from the grand coalition is harmful.

5. Identification of pivotal player & Distribution of Shared Revenue

5.1. Identification of pivotal player

A group of prosumers having some surplus energy is formed. It is worth noting that, this group includes the prosumers having surplus energy, but individually, their surplus energy is not enough to facilitate even a single consumer. Such prosumers contributions may vary in a shared contract. Therefore, it is essential to identify and classify the prosumers, based on the level of their contributions for a fair distribution of shared revenue. In order to identify the pivotal player in the grand coalitions, the Shapely-Shubik power index is employed. N prosumers are allowed to join the grand coalition. The number of sequential coalitions with N prosumers can be found by taking $N!$ The Shapley-Shubik power index of pivotal prosumer (P_i) can be found as $\sigma_i = \frac{SS_i}{N!}$, Where σ_i is the Shapley-Shubik power index of P_i, $SS_i = No.$ of sequence coalitions where P_i is pivotal and N = Number of prosumers who involve in grand coalition. After identifying the pivotal prosumer, the next objective is to distribute the shared revenue in a uniform manner. For this purpose, the proposed method employs Shapley-Value distribution. Moreover, the proposed cooperative game consists of a set N (prosumers) and a function ϑ that maps subsets of players to the real number ϑ: $2^N \to R$, with $\vartheta(\phi) = 0$, where ϕ represents the empty set. The function ϑ is called a characteristic function. The function ϑ presents the following meaning: if S is the grand coalition of the players, then $\vartheta(S)$ is called the worth of coalition S, which describes the total expected sum of payoffs the members of a S can obtain by a grand coalition. The Shapley value is a way to distribute the total shared revenue to the prosumers collaborating in grand coalition.

$$\rho_i(\vartheta) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(N - |S| - 1)!}{N!} (\vartheta(S \cup \{i\} - \vartheta(S))) \tag{19}$$

Where N is the total number of prosumers and the sum extends over all subsets S of N not containing prosumer P_i

6. Simulation and results

This section presents the experiment result of the proposed method for trading electricity in the P2P electricity market. The experiments are based on the IEEE 14-bus system, which includes consumer load consumption along with import and export prices as used in [35]. The eagerness factor δ of the consumers was randomly selected between 0 and 1 as no dataset was available. This dataset is sampled on a 30-minutes time slot, for one year, i.e. from July 2012 to June 2013. While generation of the prosumers is taken between the [0.4 kWh - 14 kWh]. The dataset uses the original number of 11 loads as shown in Fig. 3. Bus 1 represents the upstream connection to the grid, where the generator assumes unbounded power. In the case of demand-supply imbalance, energy can be traded with the grid. At any interval t the pricing signal p_{market} from the system is derived from the average of import and export market price. Prosumers and consumers are supposed to submit the $P_{propose}$, P_{offer} according to the condition described in 13 and 14 respectively. To demonstrate the working of the proposed model, demand of energy (Q_{demand}) of the consumers, eagerness factor (δ) and offered prices (p_{offer}) are taken from the consumption profile of the consumers at random time slot at $t = 00{:}\ 00{:}\ 00$. The grid import rate is taken from the price dataset as $p_{market_{import}} = 6.2\ c/KWh$

$$Q_{demand} = \{3.57, 53.09, 5.04, 0.64, 3.98, 2.62, 1.05, 0.4, 1.9, 1.48, 1.93\} \tag{20}$$

$$\delta = \{0.5, 1, 0.3, 0.6, 0.7, 0.8, 0.35, 0.1, 0.9, 0.2, 0.4\} \tag{21}$$

$$P_{offer} = \{5.42, 5.72, 5.32, 5.47, 5.47, 5.6, 5.37, 5.2, 5.68, 5.3, 5.40\} \tag{22}$$

Similarly, prosumer's surplus energy $(Q_{surplus})$ along with the $p_{propose}$ w.r.t. the $p_{market_{export}} = 4.2c/KWh$ is taken as:

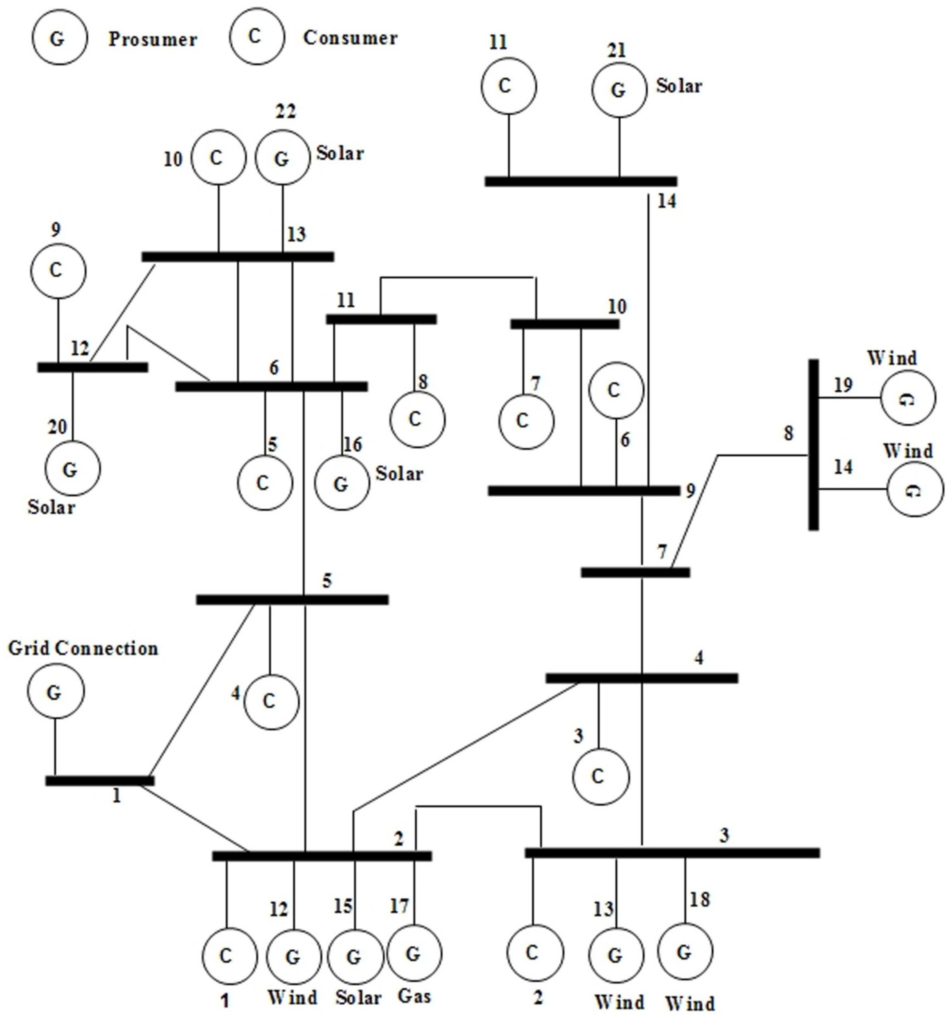


Fig. 3. IEEE 14 Bus System Model

$$Q_{surplus} = \{2.81,\ 1.46,\ 3.85,\ 1.05,\ 2.93,\ 5.99,\ 8.04,\ 3.77,\ 0.49,\ 14.52,\ 4.33\} \tag{23}$$

$$p_{propose} = \{4.93,\ 5.10,\ 5.08,\ 5.12,\ 5.03,\ 4.98,\ 4.97,\ 4.95,\ 5.08,\ 4.96,\ 5.12\} \tag{24}$$

The proposed system shares the above information except, eagerness factor among prosumers and consumers. Since all prosumers are rational and want to trade with a consumer with higher price. However, on the basis of eagerness factor (δ)consumer can be categorized in two main types (i) price-conscious consumer (δ = [0.1, 0.3]),(ii) quantity conscious consumer (δ = [0.4, 1.0]).Based on this discussion the preference lists for all consumers and prosumers are given in Table 1. These preference lists are sent back to the system following Algorithm 1 and Algorithm 2, the system will form the non-cooperative game among consumers and prosumers as: By observing the preference list of the c_1, system first makes a temporary contract between c_1 and P_{10}. After observing the c_2 list, the system finds that c_2 also gives higher priority to the prosumer P_{10}. Moreover, preference list of the prosumer P_{10}, which is based on the expected utility of the prosumer system finds that expected utility of the prosumer P_{10} will be high, if a contract is established between prosumer P_{10} and consumer c_2. Therefore, temporary contract between the consumer c_2 and prosumer P_{10} will be rejected and consumer c_1 is free and a contract between consumer c_2 and prosumer P_{10} will be formed. Furthermore, the system will again form a temporary contract between the consumer c_1 and prosumer P_8 based on the preference list of the consumer c_1. By evaluating the list of a consumer c_3, prosumer P_9 will be assigned to c_3. Next, the system examines the list of a consumer c_4 and finds that c_4 wants to make contract with the prosumer P_{10}, but prosumer P_{10} is already engaged with the consumer c_2 therefore this contract is not possible and the next potential candidate in the list of c_4 is P_8, but P_8 is engaged in the temporary contract with the c_1. Based on the list of the prosumer P_8 system finds

Table 1
Preferences lists of consumers and prosumers

C_{id}	δ	Consumer List	P_{id}	Prosumer's List
1	0.5	{10,8,7,6,11,3,5,1,2,4,9}	1	{2,9,6,4,5,1,11,7,3,10,8}
2	1	{10,8,7,6,11,3,5,1,2,4,9}	2	{2,9,6,4,5,1,11,7,3,10,8}
3	0.3	{9,4,2,1,5,3,11,6,7,8,10}	3	{2,9,6,4,5,1,11,7,3,10,8}
4	0.6	{10,8,7,6,11,3,5,1,2,4,9}	4	{2,9,6,4,5,1,11,7,3,10,8}
5	0.7	{10,8,7,6,11,3,5,1,2,4,9}	5	{2,9,6,4,5,1,11,7,3,10,8}
6	0.8	{10,8,7,6,11,3,5,1,2,4,9}	6	{2,9,6,4,5,1,11,7,3,10,8}
7	0.35	{9,4,2,1,5,3,11,6,7,8,10}	7	{2,9,6,4,5,1,11,7,3,10,8}
8	0.1	{9,4,2,1,5,3,11,6,7,8,10}	8	{2,9,6,4,5,1,11,7,3,10,8}
9	0.9	{10,8,7,6,11,3,5,1,2,4,9}	9	{2,9,6,4,5,1,11,7,3,10,8}
10	0.2	{9,4,2,1,5,3,11,6,7,8,10}	10	{2,9,6,4,5,1,11,7,3,10,8}
11	0.4	{10,8,7,6,11,3,5,1,2,4,9}	11	{2,9,6,4,5,1,11,7,3,10,8}

Table 2
1st contract formulation using Algorithm 1

c_i	p_i	$\varphi_{c,P}$	Demand	Surplus	Demand Left	Surplus Left	Trading Price
1	~~10, 8, 7, 6, 11~~,3	1,3	3.5705	3.8523	0	0.2818	5.2548
2	10	2,10	53.0915	14.5260	43.0915	4.5260	5.3439
3	~~9~~,4	3,4	5.0461	1.0569	3.9892	0	5.2232
4	~~10,8,7~~,6	4,6	0.6454	5.9997	0	5.3543	5.2293
5	~~10,8,7,6~~,11	5,11	3.9833	4.3376	0	0.3542	5.2980
6	~~10,8~~,7	6,7	2.6213	8.0486	0	5.4273	5.2904
7	9	7,9	1.0599	0.4947	0.5651	0	5.2298
8	~~9,4,1,2~~,5	8,5	0.4073	2.9378	0	2.5305	5.1210
9	~~10~~,8	9,8	1.9098	13.7766	0	11.8667	5.3191
10	~~9,4,2~~,1	10,1	1.4878	2.8147	0	1.3268	5.3089
11	~~10,8,7,6,11,3~~,2	11,2	1.9380	1.4668	0.4711	0	5.2561

that P_8 prefers to establish contract with the consumer c_4 rather than c_1. Therefore, system rejects the temporary contract between P_8 and c_1 and forms a new temporary contract between P_8 and c_4. The process will continue until all consumers and prosumers are bonded in a contract. Table 2 provides the information about all non-cooperative contract among consumers and prosumers along with the values of energy trading. As prosumer or consumer cannot trade energy more than $\xi = (10KWh)$ in a single contract and any prosumer or consumer cannot repeat their contract until condition described in (9) and (12) are fulfilled. According to the left surplus energy of prosumers and demand of consumers at the end of first contract system will formulate the new contract $\varphi'_{consumer,prosumer}$ by following Algorithm 2. Moreover, table 3 shows that one consumer c_2 is left, that still requires an energy demand of 33.09 KWh. Furthermore, some prosumers are still left at the end of algorithm-(1&2), which still have surplus energy but none of the prosumer, can cater the demand of the consumer c_2 Therefore, prosumer can export their surplus energy to the grid or they can offer a contract to the consumer c_2 The revenue a prosumer can earn by exporting surplus towards the grid will be low as compared to the revenue they can earn by selling surplus to the consumer c_2 as described in (6). Therefore, instead of making individual contracts of the consumer c_2 with the available prosumers, a grand coalition in terms of cooperative game (Stage-Hunt) is applied, the list of available prosumers along with the surplus energy is presented in Table 4. In grand coalition different prosumers participate with varying surplus energy. Shapely-Shubik power indexes σ_i of the prosumers is calculated with $\sigma_i = \frac{SS_i}{N!}$ and given in Table 4, it is clear that every prosumer has important role in the grand coalitions as total surplus (16.6421 KWh) is less than the total demand (33.0915 KWh). However, in this case the marginal contribution of each prosumer towards the grand coalition, prosumer P_7 is the pivotal player and the revenue earned by the grand coalition is distributed by following the shapely value. The energy required by the consumer c_2 still remains i.e. (33.0915-16.6421 = 16.4494), as there is no prosumer left in the market to which c_2 can trade with, therefore the remaining required energy will be imported from the grid. In Figs. 3 and 4, we present the simulation results of our proposed model and display some randomly selected prosumers' revenue functions considering the traditional market revenues that a prosumer gets after exporting the energy to the grid. It can be seen from the Fig. 4 (a-b) that the proposed method increases the revenues of the prosumers. Variation in the revenue values of different prosumers is due to the fact that at different time slots, prosumers are contributing different amounts of surplus electricity in P2P electricity market at different prices. Fig. 4 (c and d) represents the comparison of consumer's electricity bill of proposed model (which includes P2P electricity market bill, Grid Charges), with traditional electricity import price as baseline. The proposed method attempts to increase the revenues of the prosumers by introducing a platform through, which they can easily and effectively trade their surplus electricity with the consumers rather than exporting it to the grid. The proposed system can increase revenues as shown in Fig. 4 (a-b).Fig. 4 (e-f) shows the saving in electricity bill of all the consumers and revenue earned of all the prosumers over the one-year simulated data respectively. Also, our proposed model reduces the amount of electricity exported to the grid which is in the favor of prosumers as shown in Fig. 4 (g). The proposed model also makes sure that the quantity of electricity imported from the grid is reduced which results in a reduction in consumer's electricity bills. The amount of electricity imported from the grid is presented in Fig. 4 (h).

Table 3
Contract formulation using Algorithm 2

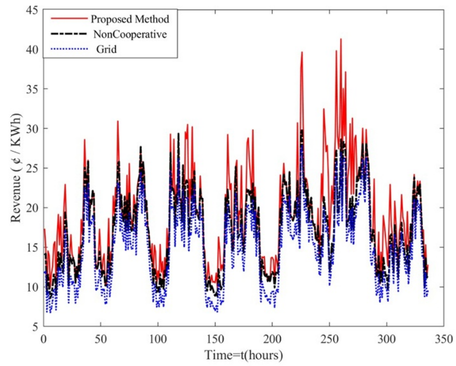
$\varphi'_{c,P}$	Demand	Surplus	Demand Left	Surplus Left	Trading Price
$\varphi'_{2,8}$	43.091	11.8667	33.0915	1.8667	5.3387
$\varphi'_{3,6}$	3.9892	5.3543	0	1.3651	5.1523
$\varphi'_{7,1}$	0.5651	1.3268	0	0.7617	5.1566
$\varphi'_{11,10}$	0.4711	4.5260	0	4.0548	5.1860

Table 4
Shapley Shublik power index

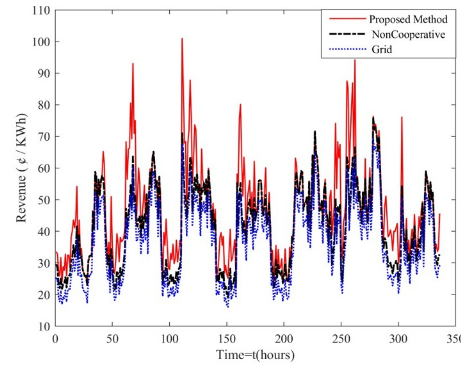
P_i	Shapley Shublik power index σ_i	Surplus left(KWh)
1	0.042857	0.7617
3	0.016667	0.2818
5	0.135714	2.5305
6	0.085714	1.3651
7	0.373810	5.4273
8	0.109524	1.8667
10	0.219048	4.0548
11	0.016667	0.3542
		Total = 16.6421

7. Conclusion

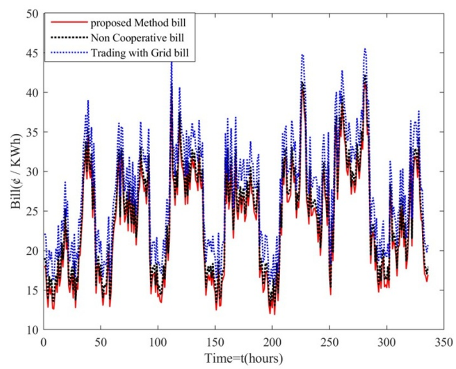
In this paper, a novel model for contract formulation in P2P electricity market is proposed, which aims to ensure the stability of contract, enhancing the utilities for both prosumers and consumers and also encouraging them to participation in P2P electricity market. The proposed model enables every prosumer and consumer to establish a contract based on their own preference. Through theoretical analysis, this paper presented a detailed analysis of the prosumers' and consumers' perspectives at the same time. It is a win-win situation for both the prosumer and consumer. Additionally, the proposed model



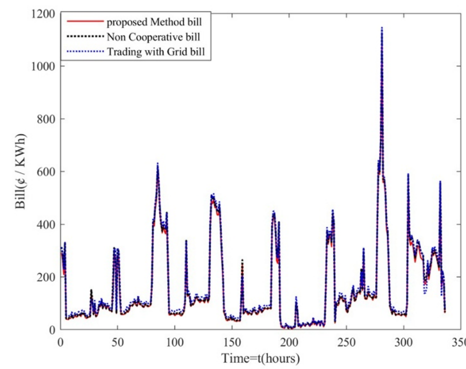
(a) Revenue of P_1



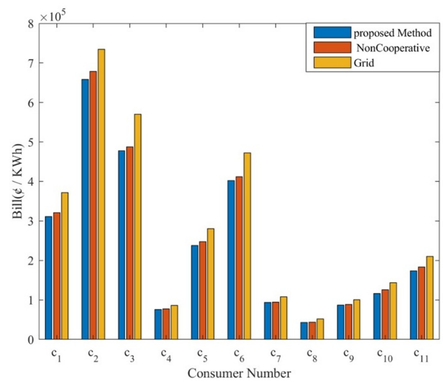
(b) Revenue of P_6



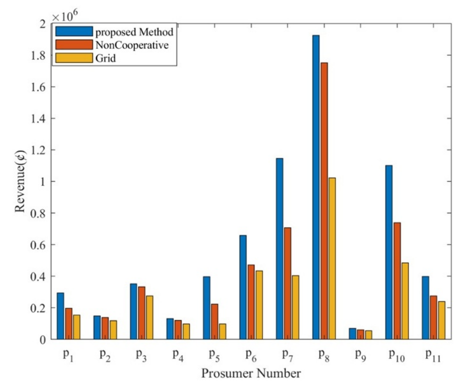
(c) Bill of c_1



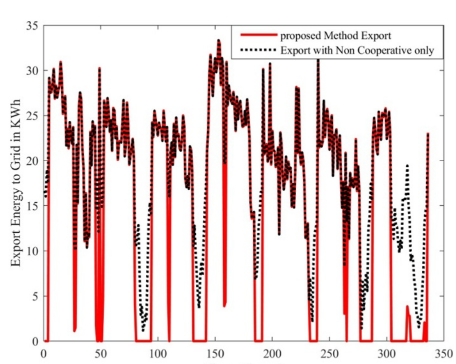
(d) Bill of c_2



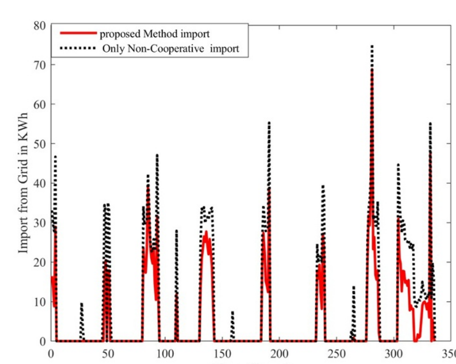
(e) Energy bill of all consumers spanning over one year data



(f) Revenue of all prosumers spanning over one year data



(g) Energy Export in KWh



(h) Energy Import in KWh

Fig. 4. Electricity buy/sell in market

considers the prosumers with smaller amount of surplus energy and are near to market exit.

CRediT authorship contribution statement

Waqas Amin: Conceptualization, Methodology, Visualization, Investigation, Software, Writing - original draft, Writing - review & editing. **Qi Huang:** Writing - original draft, Writing - review & editing, Supervision. **M. Afzal:** Visualization, Writing - review & editing. **Abdullah Aman Khan:** Writing - review & editing. **Khalid Umer:** Conceptualization, Methodology, Writing - review & editing. **Syed Adrees Ahmed:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Y. Luo, S. Itaya, S. Nakamura, P. Davis, Autonomous cooperative energy trading between prosumers for microgrid systems, 39th Annual IEEE Conference on Local Computer Networks Workshops, IEEE, 2014, pp. 693–696.
[2] C. Bussar, P. Stöcker, Z. Cai, L. Moraes Jr, D. Magnor, P. Wiernes, N. van Bracht, A. Moser, D.U. Sauer, Large-scale integration of renewable energies and impact on storage demand in a european renewable power system of 2050ìsensitivity study, J. Energy Storage 6 (2016) 1–10.
[3] Investopedia, Jim Chappelow, Peer-to-Peer (P2P) economy, (2018). https://www.investopedia.com/terms/p/peertopeer-p2p-economy.asp (accessed 03 March 2020).
[4] J. Hamari, M. Sjöklint, A. Ukkonen, The sharing economy: Why people participate in collaborative consumption, Journal of the association for information science and technology 67 (9) (2016) 2047–2059.
[5] CISION PR Newswire, sonnen GmbH, sonnencommunity connects households and makes conventional electricity suppliers obsolete through self-generated power, (2015). https://www.prnewswire.com/news-releases/sonnencommunity-connects-households-and-makes-conventional-electricity-suppliers-obsolete-through-self-generated-power-300184496.html (accessed 03 March 2020).
[6] good energy, Piclo, Shaping the future of energy with the Piclo trial, (2015). https://www.goodenergy.co.uk/blog/2015/03/10/shaping-the-future-of-energy-with-the-piclo-trial-100315/ (accessed 03 March 2020).
[7] Ena, John Bradley, Electricity network transformation roadmap (Network Business Models), (2015). https://www.energynetworks.com.au/assets/uploads/network_business_models_accenture_report_john_bradley_final.pdf (accessed 03 March 2020).
[8] T. Liu, X. Tan, B. Sun, Y. Wu, X. Guan, D.H. Tsang, Energy management of cooperative microgrids with p2p energy sharing in distribution networks, 2015 IEEE international conference on smart grid communications (SmartGridComm), IEEE, 2015, pp. 410–415.
[9] M.R. Alam, M. St-Hilaire, T. Kunz, An optimal p2p energy trading model for smart homes in the smart grid, Energy Effi. 10 (6) (2017) 1475–1493.
[10] N. Liu, X. Yu, C. Wang, C. Li, L. Ma, J. Lei, Energy-sharing model with price-based demand response for microgrids of peer-to-peer prosumers, IEEE Trans. Power Syst. 32 (5) (2017) 3569–3583.
[11] R. Roche, B. Blunier, A. Miraoui, V. Hilaire, A. Koukam, Multi-agent systems for grid energy management: A short review, IECON 2010-36th Annual Conference on IEEE Industrial Electronics Society, IEEE, 2010, pp. 3341–3346.
[12] A. Eydeland, K. Wolyniec, Energy and power risk management: New developments in modeling, pricing, and hedging, 206 John Wiley & Sons, 2003.
[13] F.A. Wolak, An empirical analysis of the impact of hedge contracts on bidding behavior in a competitive electricity market, Int. Econ. J. 14 (2) (2000) 1–39.
[14] C.D. Wolfram, Measuring duopoly power in the british electricity spot market, Am. Econ. Rev. 89 (4) (1999) 805–826.
[15] Y. Dai, H. Yang, J. Wu, Electricity supply chain coordination based on quantity discount contracts, 2009 IEEE International Conference on Industrial Technology, IEEE, 2009, pp. 1–5.
[16] A. Nagurney, Z. Liu, M.-G. Cojocaru, P. Daniele, Dynamic electric power supply chains and transportation networks: An evolutionary variational inequality formulation, Transport. Res. Part E 43 (5) (2007) 624–646.
[17] K. Wu, A. Nagurney, Z. Liu, J.K. Stranlund, Modeling generator power plant portfolios and pollution taxes in electric power supply chain networks: A transportation network equilibrium transformation, Transport. Res. Part D 11 (3) (2006) 171–190.
[18] M. Motalleb, R. Ghorbani, Non-cooperative game-theoretic model of demand response aggregator competition for selling stored energy in storage devices, Appl. Energy 202 (2017) 581–596.
[19] R. Duan, G. Deconinck, Future electricity market interoperability of a multi-agent model of the smart grid, 2010 International Conference on Networking, Sensing and Control (ICNSC), IEEE, 2010, pp. 625–630.
[20] M. Mashhour, M. Golkar, S. Moghaddas-Tafreshi, Extending market activities for a distribution company in hourly-ahead energy and reserve markets–part i: Problem formulation, Energy Conver. Manag. 52 (1) (2011) 477–486.
[21] D.J. Vergados, I. Mamounakis, P. Makris, E. Varvarigos, Prosumer clustering into virtual microgrids for cost reduction in renewable energy trading markets, Sustain. Energy Grids Netw. 7 (2016) 90–103.
[22] M. Marzband, M. Javadi, J.L. Domínguez-García, M.M. Moghaddam, Non-cooperative game theory based energy management systems for energy district in the retail market considering der uncertainties, IET Gener., Trans. Distrib. 10 (12) (2016) 2999–3009.
[23] K. Ma, C. Wang, J. Yang, C. Hua, X. Guan, Pricing mechanism with noncooperative game and revenue sharing contract in electricity market, IEEE Trans. Cybernet. (99) (2017) 1–10.
[24] D. Srinivasan, L.T. Trung, C. Singh, Bidding and cooperation strategies for electricity buyers in power markets, IEEE Syst. J. 10 (2) (2014) 422–433.
[25] M.J.V. Pakdel, S. Ghaemi, B. Mohammadi-Ivatloo, J. Salehi, P. Siano, Modeling non-cooperative game of gencos' participation in electricity markets with prospect theory, IEEE Trans. Ind. Inform. (2019).
[26] H. Wang, T. Huang, X. Liao, H. Abu-Rub, G. Chen, Reinforcement learning for constrained energy trading games with incomplete information, IEEE Trans. Cybernet. 47 (10) (2016) 3404–3416.
[27] K. Ma, G. Hu, C.J. Spanos, A cooperative demand response scheme using punishment mechanism and application to industrial refrigerated warehouses, IEEE Trans. Ind. Inform. 11 (6) (2015) 1520–1531.
[28] B. Zhang, C. Jiang, J.-L. Yu, Z. Han, A contract game for direct energy trading in smart grid, IEEE Trans. Smart Grid 9 (4) (2016) 2873–2884.
[29] F. Moret, P. Pinson, Energy collectives: a community and fairness based approach to future electricity markets, IEEE Trans. Power Syst. (2018).
[30] A.C.Z. de Souza, N. De Nadai, B. Portelinha, F.M. Marujo D., D.Q. Oliveira, Microgrids Operation in Islanded Mode, Sustainable Development in Energy Systems Springer, Cham, 2017.
[31] W. Tushar, C. Yuen, W. Li, D.B. Smith, T. Saha, K.L. Wood, Motivational psychology driven AC management scheme: A responsive design approach, IEEE Trans. Comput. Social Syst. 5 (1) (2018) 289–301, https://doi.org/10.1109/TCSS.2017.2788922.
[32] W. Tushar, B. Chai, C. Yuen, D.B. Smith, H.V. Poor, Energy management for a user interactive smart community: A stackelberg game approach, 2014 IEEE Innovative Smart Grid Technologies - Asia (ISGT ASIA), (2014), pp. 709–714, https://doi.org/10.1109/ISGT-Asia.2014.6873879.
[33] M. Afzal, K. Umer, W. Amin, M. Naeem, D. Cai, Z. Zhenyuan, Q. Huang, Blockchain based domestic appliances scheduling in community microgrids, 2019 IEEE Innovative Smart Grid Technologies - Asia (ISGT Asia), (2019), pp. 2842–2847, https://doi.org/10.1109/ISGT-Asia.2019.8881074.
[34] J. Wang, Q. Wang, N. Zhou, Y. Chi, A novel electricity transaction mode of microgrids based on blockchain and continuous double auction, Energies 10 (12) (2017) 1971.
[35] T. Sousa, T. Soares, P. Pinson, F. Moret, T. Baroche, E. Sorin, Peer-to-peer and community-based markets: acomprehensive review, Renew. Sustain. Energy Rev. 104 (2019) 367–378.
[36] M. Afzal, Q. Huang, W. Amin, K. Umer, A. Raza, M. Naeem, Blockchain Enabled Distributed Demand Side Management in Community Energy System With Smart Homes, IEEE Access 8 (2020) 37428–37439, https://doi.org/10.1109/ACCESS.2020.2975233.

Waqas Amin received his BS and MS degree in Electrical Engineering from University of Lahore(UOL), Pakistan in 2010 and 2016 respectively. He served as a lecturer in University of Lahore from 2010-2017. Currently, he is pursuing PhD degree with the Sichuan State Provincial Laboratory of Power System Wide-Area Measurement and Control, School of Mechanical and Electrical Engineering, University of Electronic Science and Technology of China, Chengdu, China.His current research interests include energy management, distributed electricity trading, and blockchain technology in power system.

Huang Qi (S99, M03, SM09) was born in Guizhou province in the People's Republic of China. He received his BS degree in Electrical Engineering from Fuzhou University in 1996, MS degree from Tsinghua University in 1999, and Ph.D degree from Arizona State University in 2003. He is currently a professor at University of Electronics Science and Technology of China (UESTC), the Executive Dean of School of Mechanical and Electrical Engineering, UESTC, and the director of Sichuan State Provincial Lab of Power System Wide-area Measurement and Control. He is a member of IEEE since 1999. His current research and academic interests include power system instrumentation, power system monitoring and control, power system high performance computing and Power Market.

Muhammad Afzal received the BS in Computer Engineering in 2010 and MS in Electrical Engineering in 2016 from COMSATS University Islamabad, Wah cantt, Pakistan. He is currently pursuing the Ph.D. degree with the Sichuan State Provincial Laboratory of Power System Wide-Area Measurement and Control, School of Mechanical and Electrical Engineering, University of Electronic Science and Technology of China, Chengdu, China. His current research interests include optimization of energy managment, energy informatics, power market, and blockchain technology in power system.

Abdullah Aman Khan received his master's degree in the field of Computer Engineering form National University of Science and Technology (NUST), Punjab, Pakistan in 2014. He is currently perusing PhD degree in the field of computer science and technology form the school of computer science and engineering, University of Electronic Science and Technology, Sichuan, Chengdu, P.R. China. His main area of research includes electronics design, machine vision and intelligent systems.

Khalid Umer received the BS in Electrical Engineering from University of Engineering and Technology (UET) Lahore and MS in Electronic Science and Technology from University of Electronic Science and Technology of China (UESTC)in 2013 and 2016 respectively. He is currently pursuing the Ph.D. degree with the Sichuan State Provincial Laboratory of Power System Wide-Area Measurement and Control, School of Mechanical and Electrical Engineering, UESTC. His current research interests include energy management, Blockchain, P2P energy trading, Game theory and distributed optimization.

Syed Adrees Ahmed received his BS degree in Electrical Engineering from COMSATS University Pakistan in 2016 and MS degree in Control Science and Engineering in 2019, with study on Optimal Energy trading in P2P power market, from University of Electronic Science and Technology of China(UESTC). He is currently pursuing PhD degree from the same University (UESTC). His current research interests include energy managment, distributed electricity trading, Blockchain, P2P Energy trading and Game theory and Optimization.